

DUOIDAL LINEAR SPECIES, FACTORIZATION LOCALITY, AND BAR DENOMINATOR PRODUCTS

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ABSTRACT. The Cauchy and Hadamard products on linear species admit canonical braided duoidal structures in both orientations. Their interchange maps are diagonal inclusion and off-diagonal-killing projection, and $\iota\pi$ is the idempotent onto the common-decomposition summand. These structures are globally non-strong. Independently, a factorization right D -module on the reduced Ran space carries collision descent and coherent factorization equivalences. An ambient augmented E_1 -structure is factorization-local precisely when its entire operadic action lifts through the factorization category; the usual disjointness-locus multiplication square is only a necessary binary consequence.

For a positively root-graded Lie superalgebra satisfying a finite-factorization condition, the completed total Euler supercharacter of the normalized algebraic bar of $U(\mathfrak{n}_+)$ is the signed root product. Under the precise hypotheses of a superdenominator theorem this is its product factor. For the Gritsenko–Nikulin algebra, a shifted formal Fourier map followed by a normally convergent cusp realization gives $D_5(2Z) = 64^{-1}\Delta_5(2Z)$. Weightwise duality with Goncharova’s continuous cohomology gives the two positive-Witt homology classes in weights $(3n^2 \pm n)/2$. The Motzkin–Riordan relation is proved independently by its generating functions. A comparison between the factorization bar and the root-graded algebraic bar is stated conditionally; no such comparison is used in the proved results.

I. THE COMPARISON PROBLEM

Dividing a set of insertions into two groups gives an ordered decomposition. Two operations can use the same decomposition or choose their decompositions independently. For a singleton there are two ordered decompositions, so these choices give two and four possibilities, respectively. Inclusion of the common choices and projection onto them give a retraction. They do not give an invertible exchange.

Linear species assign vector spaces to finite sets and transport them along bijections. The Cauchy product distributes a set between its two inputs. The Hadamard product evaluates both inputs on the whole set. Their interchange maps implement the preceding inclusion and projection. The resulting two duoidal structures retain both products and their coherent compatibility, with different units. Noninvertibility is part of this construction and leaves its bimonoids well defined.

When insertion points lie on a curve, they can also collide. A factorization right D -module specifies coefficients on configurations, their restriction to collision diagonals, and their decomposition away from those diagonals. An ordered product on it must preserve all these maps and their higher coherences. Such a product defines an internal simplicial bar. Its realization in the factorization category requires the colimit hypotheses of Theorem 5.2.

A second calculation starts from root-graded Lie superalgebras. The normalized bar of their enveloping algebras has an Euler supercharacter determined by the even and odd root spaces. Finite factorization of weights makes each coefficient a finite calculation. The resulting product has a precise Igusa specialization, including a Weyl shift and an analytic domain of convergence. To obtain that character from a factorization bar, one still needs a map between the two chain constructions. Problem 6.4 specifies its coefficients and operations, and Theorem 6.5 gives the consequence once those data are constructed.

2. CONVENTIONS

Throughout, k is a field of characteristic zero. The groupoid $\text{Fin}^\approx$ contains all finite sets, including $\emptyset$, and species are covariant under bijections. By contrast, Fin^{surj} denotes the category of nonempty finite sets and surjections. It indexes the reduced Ran space; there is no empty Ran configuration in this convention.

Super vector spaces are $\mathbb{Z}/2$ -graded and their morphisms preserve intrinsic parity. Mapping complexes, dg objects, and A_∞ -algebras use cohomological degrees: the differential has degree $+1$, $(A[d])^p = A^{p+d}$, and m_n has degree $2 - n$. Normalized simplicial bars and Chevalley–Eilenberg chains carry a separate nonnegative homological degree r , whose differential lowers r by one. In a total Euler supercharacter a class of this homological degree and intrinsic parity ϵ has sign $(-1)^{r+\epsilon}$. These two grading conventions are not identified. When a cohomological degree p and a bar degree r occur together, the cohomological totalization has degree $p - r$.

For the geometric statements, X is a smooth separated complex algebraic curve. All categories of D -modules are stable ∞ -categories of right D -modules. Extraordinary pullback is denoted by $(-)^!$, and every collision, factorization, and refinement equivalence is understood homotopy coherently.

3. THE TWO DUOIDAL STRUCTURES ON LINEAR SPECIES

Let

$$\text{Sp}_k = \text{Fun}(\text{Fin}^\approx, \text{Vect}_k).$$

For $F, G \in \text{Sp}_k$, define the Cauchy and Hadamard products by

$$(F \boxtimes_{\text{C}} G)(I) = \bigoplus_{I=I_1 \sqcup I_2} F(I_1) \otimes G(I_2), \quad (F \boxtimes_{\text{H}} G)(I) = F(I) \otimes G(I).$$

The decompositions in the first sum are ordered. Write $\mathbb{U}$ for the Cauchy unit and $\mathbb{E}$ for the Hadamard unit:

$$\mathbb{U}(\emptyset) = k, \quad \mathbb{U}(I) = 0 \quad (I \neq \emptyset), \quad \mathbb{E}(I) = k.$$

For two ordered decompositions $d = (I_1, I_2)$ and $e = (J_1, J_2)$ of I , put

$$T_{d,e} = F_1(I_1) \otimes F_2(I_2) \otimes G_1(J_1) \otimes G_2(J_2).$$

After the evident symmetry of tensor factors,

$$\begin{aligned} L_I &:= ((F_1 \boxtimes_{\text{H}} G_1) \boxtimes_{\text{C}} (F_2 \boxtimes_{\text{H}} G_2))(I) \cong \bigoplus_d T_{d,d}, \\ R_I &:= ((F_1 \boxtimes_{\text{C}} F_2) \boxtimes_{\text{H}} (G_1 \boxtimes_{\text{C}} G_2))(I) \cong \bigoplus_{d,e} T_{d,e}. \end{aligned}$$

Diagonal inclusion and projection define natural transformations

$$\iota : L \longrightarrow R, \quad \pi : R \longrightarrow L.$$

Theorem 3.1 (Cauchy–Hadamard duoidality). *The transformations ι and π define braided duoidal structures in the two respective orientations*

$$(\text{Sp}_k, \boxtimes_{\text{C}}, \mathbb{U}, \boxtimes_{\text{H}}, \mathbb{E}; \iota), \quad (\text{Sp}_k, \boxtimes_{\text{H}}, \mathbb{E}, \boxtimes_{\text{C}}, \mathbb{U}; \pi).$$

For the ι -orientation the three unit morphisms are

$$\mathbb{U} \longrightarrow \mathbb{U} \boxtimes_{\text{H}} \mathbb{U}, \quad \mathbb{E} \boxtimes_{\text{C}} \mathbb{E} \longrightarrow \mathbb{E}, \quad \mathbb{U} \longrightarrow \mathbb{E}.$$

For the π -orientation they are

$$\mathbb{E} \longrightarrow \mathbb{E} \boxtimes_{\text{C}} \mathbb{E}, \quad \mathbb{U} \boxtimes_{\text{H}} \mathbb{U} \longrightarrow \mathbb{U}, \quad \mathbb{E} \longrightarrow \mathbb{U}.$$

Moreover,

$$\pi \iota = \text{id}_L, \quad e := \iota \pi, \quad e^2 = e,$$

and e has image $\bigoplus_d T_{d,d}$.

Proof. The map ι_I includes the summands indexed by $d \mapsto (d, d)$, while π_I is the identity on those summands and zero on every $T_{d,e}$ with $d \neq e$. Hence $\pi_I \iota_I = \text{id}$, and $\iota_I \pi_I$ is the displayed idempotent.

The unit maps contain information not visible in this retraction. The map $\mathbb{E} \boxtimes_C \mathbb{E} \rightarrow \mathbb{E}$ is the fold over all ordered decompositions, and $\mathbb{E} \rightarrow \mathbb{E} \boxtimes_C \mathbb{E}$ is the biproduct diagonal. The maps $\mathbb{U} \rightarrow \mathbb{E}$ and $\mathbb{E} \rightarrow \mathbb{U}$ are the identity at $\emptyset$ and zero elsewhere. The two maps involving $\mathbb{U} \boxtimes_H \mathbb{U}$ are the canonical unit identifications at $\emptyset$ and the unique maps between zero spaces elsewhere.

Every associativity diagram decomposes into matrix entries indexed by ordered decompositions into three blocks. In the ι -orientation both composites retain exactly those entries whose decomposition labels agree; in the π -orientation both composites kill exactly the complementary entries. The unit and braided diagrams reduce in the same way to the fold, diagonal, identity, or zero maps just described. This proves the coherence axioms. For vector species these two orientations are also the braided 2-monoidal structures of [1, Definitions 6.1 and 6.3, equations (8.73)–(8.74), Propositions 8.58 and 8.68]; the three-map unit datum is the one in [10, Definition 2.1]. $\square$

Proposition 3.2 (The common-decomposition summand). *For fixed F_1, F_2, G_1, G_2 and I , the map ι_I is an isomorphism if and only if every off-diagonal summand $T_{d,e}$, $d \neq e$, is zero. This is an objectwise condition; it does not define a unital monoidal subcategory containing the inherited units.*

Proof. The target is the direct sum of the image of ι_I and all off-diagonal summands. Thus ι_I is surjective exactly when the latter vanish. $\square$

Corollary 3.3 (Failure of strongness). *If $k \neq 0$, neither duoidal structure in Theorem 3.1 is strong. There is no full subcategory containing both inherited units on which all interchange and unit morphisms become isomorphisms.*

Proof. Take $I = \{*\}$ and all four inputs equal to $\mathbb{E}$. There are two ordered decompositions of I , and therefore

$$L_I \cong k^2, \quad R_I \cong k^4.$$

In compatible coordinates,

$$\iota(a, b) = (a, 0, 0, b), \quad \pi(a, b, c, d) = (a, d), \quad \iota\pi = \text{diag}(1, 0, 0, 1).$$

The interchange is not invertible. Independently, $\mathbb{U}(\{*\}) = 0$ and $\mathbb{E}(\{*\}) = k$, so neither cross-unit map is invertible. Any subcategory containing the inherited units contains this obstruction. $\square$

Remark 3.4. Bimonoids are defined for a duoidal category without requiring strong interchange. The idempotent e records a useful summand, but invertibility is neither part of duoidality nor needed for the definition of a bimonoid.

4. FACTORIZATION LOCALITY ON THE REDUCED RAN SPACE

Species record decomposition labels but have no maps for collisions of points on X . Collision diagonals supply these maps. The factorization condition then relates their restrictions to products on disjoint configurations.

For a surjection $q : I \twoheadrightarrow J$, let

$$\Delta_q : X^J \longrightarrow X^I$$

be the collision diagonal. The reduced Ran category of right D -modules is

$$\mathrm{DMod}^r(\mathrm{Ran}X) = \lim_{I \in \mathrm{Fin}^{\mathrm{surj}}} \mathrm{DMod}^r(X^I),$$

where the transition attached to q is

$$\Delta_q^! : \mathrm{DMod}^r(X^I) \longrightarrow \mathrm{DMod}^r(X^J).$$

Thus a Ran object consists of objects $A_I \in \mathrm{DMod}^r(X^I)$ and equivalences

$$\theta_q : \Delta_q^! A_I \xrightarrow{\sim} A_J,$$

with coherent identity and composition data

$$\theta_{r \circ q} \simeq \theta_r \circ \Delta_r^!(\theta_q)$$

for composable surjections.

For $\pi : I \twoheadrightarrow J$, put $I_j = \pi^{-1}(j)$ and

$$U(\pi) = \{(x_i) \in X^I : x_i \neq x_{i'} \text{ whenever } \pi(i) \neq \pi(i')\}, \quad j_\pi : U(\pi) \hookrightarrow X^I.$$

Definition 4.1 (Factorization right D -module). A factorization right D -module is a Ran object A together with equivalences

$$\phi_\pi : j_\pi^! A_I \xrightarrow{\sim} j_\pi^! \left(\boxtimes_{j \in J} A_{I_j} \right)$$

for every π , compatibly with collision maps, bijections, refinements of surjections, and every higher refinement simplex. Morphisms respect all θ_q , ϕ_π , and their coherences.

We write $\mathrm{FactD}(X)$ for the resulting factorization subcategory, equivalently the Francis–Gaitsgory category

$$\mathrm{FactD}(X) = \mathrm{ComCoalg}_{\mathrm{Fact}}^{\mathrm{ch}}(\mathrm{DMod}^r(\mathrm{Ran}X)).$$

It includes in the category of chiral commutative coalgebras, which in turn has a forgetful functor to $\mathrm{DMod}^r(\mathrm{Ran}X)$. The composite is not a full embedding of $\mathrm{FactD}(X)$ into underlying Ran D -modules. The reduced Ran limit, chiral convolution, and factorization condition are given in [7, Section 2.1.1, Definition 2.1.2, Lemma 2.3.4, Definitions 2.4.5 and 2.4.7]; the coherent-system interpretation is [7, Remark 2.4.8].

4.1. The pointwise tensor. For right D -modules define

$$(A \otimes^! B)_I = A_I \otimes^! B_I.$$

The collision equivalence is the composite

$$\Delta_q^!(A_I \otimes^! B_I) \simeq \Delta_q^! A_I \otimes^! \Delta_q^! B_I \xrightarrow{\theta_q^A \otimes \theta_q^B} A_J \otimes^! B_J.$$

The factorization equivalence is

$$\begin{aligned} j_\pi^!(A_I \otimes^! B_I) &\simeq j_\pi^! A_I \otimes^! j_\pi^! B_I \\ &\xrightarrow{\phi_\pi^A \otimes \phi_\pi^B} j_\pi^! \left(\boxtimes_j A_{I_j} \right) \otimes^! j_\pi^! \left(\boxtimes_j B_{I_j} \right) \\ &\xrightarrow{\mathrm{ex}} j_\pi^! \left(\boxtimes_j (A_{I_j} \otimes^! B_{I_j}) \right). \end{aligned}$$

The exchange map and its coherence are part of this construction. Symmetric monoidality of extraordinary pullback and external-product exchange supplies the identity, composition, permutation, and refinement coherences. They give the symmetric monoidal category

$$(\text{FactD}(X), \otimes^!, \omega_{\text{Ran}}), \quad (\omega_{\text{Ran}})_I = \omega_{X^I},$$

and a strong symmetric monoidal forgetful functor

$$U : \text{FactD}(X) \longrightarrow \text{DMod}^r(\text{Ran}X).$$

The convention remains reduced: there is no component $A_\emptyset$. An augmentation is a morphism $A \rightarrow \omega_{\text{Ran}}$ for the pointwise tensor and does not adjoin an empty configuration to the chiral coalgebra.

Definition 4.2 (Ordered factorization algebra). An ordered factorization algebra is an object

$$A \in \text{Alg}_{E_1}^{\text{aug}}(\text{FactD}(X), \otimes^!).$$

Proposition 4.3 (Locality as an operadic lift). *Let $A \in \text{FactD}(X)$, and let α be an augmented E_1 -algebra structure on $U(A)$. A factorization-local structure on α is precisely a lift of maps of ∞ -operads*

$$\begin{array}{ccc} & & \text{FactD}(X)^\otimes \\ & \nearrow \tilde{\alpha} & \downarrow U^\otimes \\ (E_1^{\text{aug}})^\otimes & \xrightarrow{\alpha} & \text{DMod}^r(\text{Ran}X)^\otimes. \end{array}$$

Equivalently, multiplication, unit, augmentation, every higher operation, and all coherent homotopies lift to the mapping spaces of $\text{FactD}(X)$, with the algebra color sent to the specified object A .

Proof. An augmented E_1 -algebra is a map from the augmented associative ∞ -operad into the monoidal ∞ -category. Lifting that map along $U^\otimes$ is exactly the definition of an augmented E_1 -algebra structure on A whose underlying structure is α . $\square$

The familiar multiplication square is a necessary arity-two consequence. For an ordered bipartition $d = (I_1, I_2)$ with $I_1, I_2 \neq \emptyset$, equivalently a surjection $I \twoheadrightarrow \{1, 2\}$, define

$$\begin{aligned} \lambda_d^{(2)} : j_d^!(A_I \otimes^! A_I) &\simeq j_d^! A_I \otimes^! j_d^! A_I \\ &\xrightarrow{\phi_d \otimes \phi_d} j_d^!(A_{I_1} \boxtimes A_{I_2}) \otimes^! j_d^!(A_{I_1} \boxtimes A_{I_2}) \\ &\xrightarrow{\text{ex}} j_d^!((A_{I_1} \otimes^! A_{I_1}) \boxtimes (A_{I_2} \otimes^! A_{I_2})). \end{aligned}$$

Every lift in Proposition 4.3 makes

$$\begin{array}{ccc} j_d^!(A_I \otimes^! A_I) & \xrightarrow{j_d^! m_{2,I}} & j_d^! A_I \\ \downarrow \lambda_d^{(2)} & & \downarrow \phi_d \\ j_d^!((A_{I_1} \otimes^! A_{I_1}) \boxtimes (A_{I_2} \otimes^! A_{I_2})) & \xrightarrow{j_d^!(m_{2,I_1} \boxtimes m_{2,I_2})} & j_d^!(A_{I_1} \boxtimes A_{I_2}) \end{array}$$

commute. It also makes every collision square commute:

$$\begin{array}{ccc} \Delta_q^!(A_I \otimes^! A_I) & \xrightarrow{\Delta_q^! m_{2,I}} & \Delta_q^! A_I \\ \downarrow \simeq & & \downarrow \theta_q \\ A_J \otimes^! A_J & \xrightarrow{m_{2,J}} & A_J. \end{array}$$

The left vertical map uses the monoidality of $\Delta_q^!$, followed by $\theta_q \otimes \theta_q$.

After representing the E_1 -structure by a cohomologically graded A_∞ -algebra, analogous diagrams are required for every

$$m_n : A^{\otimes n} \longrightarrow A[2-n],$$

as well as for the unit, augmentation, associahedral homotopies, arbitrary multiblock surjections, refinements, and permutations.

Example 4.4 (Binary compatibility is not sufficient). Assume $\text{char } k \neq 2$. In $\text{Fun}(BC_2, \text{Ch}_k)$, let

$$A^0 = k1 \oplus kx, \quad A^{-1} = ky,$$

with zero differential and square-zero augmentation ideal for m_2 . Give A the strictly unital minimal A_∞ -operation

$$m_3(x, x, x) = y$$

and no other nonzero higher operation. Let the involution fix $1, x$ and send y to $-y$. Multiplication, unit, and augmentation are equivariant, but m_3 is not. Thus binary equivariance does not imply descent of an E_1 -structure.

Remark 4.5 (The species and Ran carriers). The transformations ι, π take values in one fixed additive category. The components of a Ran object lie in the varying categories $\text{DMod}^r(X^I)$ and are joined by extraordinary pullback, open restriction, and collision descent. A geometric analogue of ι or π therefore requires mixed pull–push functors, projection-formula maps, Beck–Chevalley compatibilities, collision squares, units, and three-block refinement coherence.

There is also a first-arity discrepancy. Reduced binary chiral convolution has no singleton summand, since there is no surjection $\{*\} \twoheadrightarrow \{1, 2\}$. The Cauchy product of species has two singleton summands arising from empty blocks. Agreement of labels on a fixed separated stratum consequently supplies neither a Ran natural transformation nor a retraction.

5. THE INTERNAL BAR AND ITS REALIZATION

An augmented ordered product gives operations for composing adjacent insertions and for removing either endpoint. Their compatibility is exactly the simplicial structure used by the bar construction. Because the product is factorization-local, these operations remain morphisms of factorization objects before any realization is taken.

Proposition 5.1 (The internal simplicial bar). *Let $\mathcal{F} = (\text{FactD}(X), \otimes^!, \omega_{\text{Ran}})$, and let $A \in \text{Alg}_{E_1}^{\text{aug}}(\mathcal{F})$. Then*

$$\text{Bar}_r^{\mathcal{F}}(\omega_{\text{Ran}}, A, \omega_{\text{Ran}}) = \omega_{\text{Ran}} \otimes^! A^{\otimes^! r} \otimes^! \omega_{\text{Ran}}$$

defines a functor $\Delta^{\circ p} \rightarrow \mathcal{F}$. The inner faces are induced by multiplication, the two outer faces by the augmentation, and the degeneracies by the unit.

Proof. Every simplicial operator is a composite of tensor products of morphisms in $\mathcal{F}$. The simplicial identities are the coherent unital associativity identities of the augmented E_1 -algebra. $\square$

Conditional Theorem 5.2 (Realized factorization bar). *Assume the geometric realization of the simplicial object in Proposition 5.1 exists in $\mathcal{F}$, and assume the canonical map*

$$\left| U \operatorname{Bar}_{\bullet}^{\mathcal{F}}(\omega_{\operatorname{Ran}}, A, \omega_{\operatorname{Ran}}) \right|_{\operatorname{DMod}^r(\operatorname{Ran} X)} \longrightarrow U \left(\left| \operatorname{Bar}_{\bullet}^{\mathcal{F}}(\omega_{\operatorname{Ran}}, A, \omega_{\operatorname{Ran}}) \right|_{\mathcal{F}} \right)$$

is an equivalence. Then the realized object is a factorization right D -module whose underlying Ran object is the ambient realized bar. If the ambient tensor preserves geometric realizations separately, that underlying object computes

$$\omega_{\operatorname{Ran}} \otimes_{U(A)}^{\mathbb{L}} \omega_{\operatorname{Ran}}.$$

Remark 5.3 (Normalization and deconcatenation). Suppose normalized reduced chains and the necessary sums exist in a chosen additive monoidal enhancement, and first suppose that A has a strict augmented dg-associative representative. With $\tilde{A} = \operatorname{fib}(A \rightarrow \omega_{\operatorname{Ran}})$, let $s\tilde{A}$ denote a copy of $\tilde{A}$ placed in independent bar homological degree one; s is not the cohomological shift $[1]$. The reduced bar is the bigraded coalgebra

$$\overline{B}(A) = T^c(s\tilde{A}) = \bigoplus_{r \geq 0} (s\tilde{A})^{\otimes r}.$$

Its internal differential raises cohomological degree and preserves r ; its simplicial bar differential is the coderivation induced by multiplication and lowers r by one. Both have degree $+1$ after the totalization $p - r$. For a general cohomological A_{∞} -representative, the full bar coderivation instead has a component induced by every operation m_n ; the component associated with m_n changes bidegree by $(2 - n, 1 - n)$ and hence again has total degree $+1$. Consequently

$$\Delta_{\operatorname{dec}}[a_1 | \cdots | a_r] = \sum_{i=0}^r [a_1 | \cdots | a_i] \otimes [a_{i+1} | \cdots | a_r]$$

is a coassociative chain map in every word length. To obtain this map inside $\operatorname{FactD}(X)$, normalization and the required sums must be created by U , tensor must preserve them, and the resulting map must respect collision and factorization descent. These colimits are not automatic: the standard category $\operatorname{Fact}(X)$ is not presentable and lacks coproducts [7, Remark 2.4.9].

6. THE ALGEBRAIC BAR DENOMINATOR

For an enveloping algebra, antisymmetrization compares ordered bar words with Chevalley–Eilenberg chains. A positive root grading makes their Euler supercharacter computable one weight at a time. We first specify the finiteness and sign conventions needed for this calculation.

In $\operatorname{Ch}_{\geq 0}(\operatorname{sVect}_k)$, write sV for a super vector space V placed in homological degree one, with its intrinsic parity unchanged. Thus s is a homological suspension, not the cohomological shift $[1]$. On bidegrees (r, ϵ) and (r', ϵ') , the symmetry has sign $(-1)^{rr' + \epsilon\epsilon'}$, while the Euler sign of one bidegree is $(-1)^{r+\epsilon}$.

Definition 6.1 (Finite-factorization monoid). A pointed commutative monoid Q_+ has the *finite-factorization property* if, for every $\beta \in Q_+$,

$$\bigsqcup_{r \geq 0} \{(\alpha_1, \dots, \alpha_r) \in (Q_+ \setminus \{0\})^r : \alpha_1 + \cdots + \alpha_r = \beta\}$$

is finite.

This condition makes

$$\widehat{\mathbb{Z}[Q_+]} = \prod_{\beta \in Q_+} \mathbb{Z} X^{\beta}, \quad (ab)_{\beta} = \sum_{\gamma + \delta = \beta} a_{\gamma} b_{\delta},$$

a ring. It is stronger than finite-dimensionality of the homogeneous pieces. For example, the pointed monoid generated inside $\mathbb{Z}^2$ by $(1, j)$ and $(0, -j)$, $j \geq 1$, has infinitely many decompositions

$$(1, 0) = (1, j) + (0, -j).$$

Let

$$\mathfrak{n}_+ = \bigoplus_{\alpha \in Q_+ \setminus \{0\}} \mathfrak{g}_\alpha$$

be a Q_+ -graded Lie superalgebra with finite-dimensional homogeneous pieces. Give k the trivial $U(\mathfrak{n}_+)$ -module structure and define

$$\mathcal{B}(\mathfrak{n}_+) := N \text{Bar}_\bullet(k, U(\mathfrak{n}_+), k) \in \text{Ch}_{\geq 0}(\text{sVect}_k^{Q_+}).$$

In simplicial homological degree r ,

$$\mathcal{B}_r(\mathfrak{n}_+) \cong \overline{U(\mathfrak{n}_+)}^{\otimes r}.$$

Equivalently, its reduced dg-bar presentation is $T^c(\overline{sU(\mathfrak{n}_+)})$. Normalization is essential: for the augmented algebra k , the normalized bar is zero in every positive degree, whereas the unnormalized simplicial object has a copy of k in every degree.

Let

$$\widehat{\text{Ch}}_{Q_+}^{\text{pf}}(\text{sVect}_k) := \prod_{\beta \in Q_+} \text{Perf}(\text{sVect}_k)$$

be the category of Q_+ -graded supercomplexes that are bounded and finite-dimensional in each weight. Its completed convolution is

$$(C \widehat{\otimes} D)_\beta = \bigoplus_{\gamma + \delta = \beta} C_\gamma \otimes D_\delta,$$

and its unit is k in weight zero. Finite factorization makes every displayed sum finite. Define the coefficientwise Grothendieck group

$$\widehat{K}_{0,s}^{Q_+}(k) = \prod_{\beta \in Q_+} K_0(\text{Perf}(\text{sVect}_k))$$

and the homomorphism

$$\begin{aligned} \chi_{Q_+}^{\text{tot}} : \widehat{K}_{0,s}^{Q_+}(k) &\longrightarrow \widehat{\mathbb{Z}[Q_+]}, \\ (c_\beta)_\beta &\longmapsto \sum_{\beta \in Q_+} \text{sdim}(c_\beta) X^\beta. \end{aligned}$$

Here $\text{sdim}(c_\beta)$ first takes the homological Euler class and then subtracts intrinsic odd parity. For a weightwise perfect supercomplex C , write $\chi_{Q_+}^{\text{tot}}(C) := \chi_{Q_+}^{\text{tot}}([C])$ for the composite through its coefficientwise Grothendieck class. Then

$$\begin{aligned} \chi_{Q_+}^{\text{tot}}(C) &= \sum_{\beta, r} (-1)^r \text{sdim} H_r(C)_\beta X^\beta \\ &= \sum_{\beta, r, \epsilon} (-1)^{r+\epsilon} \dim H_r(C)_{\beta, \epsilon} X^\beta \in \widehat{\mathbb{Z}[Q_+]}. \end{aligned}$$

Theorem 6.2 (The algebraic bar denominator). *Under the preceding hypotheses, $\mathcal{B}(\mathfrak{n}_+)$ is weightwise perfect. Super antisymmetrization gives a natural root-graded quasi-isomorphism*

$$\text{Alt} : C_\bullet^{\text{CE}}(\mathfrak{n}_+; k) = (\text{Sym}_{\text{Ch}(\text{sVect}_k)}(\mathfrak{s}\mathfrak{n}_+), d_{\text{CE}}) \xrightarrow{\simeq} \mathcal{B}(\mathfrak{n}_+),$$

where d_{CE} lowers homological degree by one and preserves intrinsic parity. The map Alt preserves homological degree, intrinsic parity, and Q_+ -weight. Moreover,

$$\chi_{Q_+}^{\text{tot}}(\mathcal{B}(\mathfrak{n}_+)) = \prod_{\alpha \in Q_+ \setminus \{0\}} (\mathbf{1} - X^\alpha)^{\text{sdim} \mathfrak{g}_\alpha}.$$

Proof. For homogeneous $x_1, \dots, x_r$, the comparison is

$$x_1 \wedge \dots \wedge x_r \mapsto \sum_{\sigma \in S_r} \text{sgn}(\sigma) \kappa(\sigma; x) [x_{\sigma(1)} | \dots | x_{\sigma(r)}],$$

where $\kappa(\sigma; x)$ is the super Koszul sign. Compatibility with the differentials is the super antisymmetrization calculation. Filter both sides by PBW degree. In each fixed Q_+ -weight this filtration is finite and exhaustive. Super PBW gives

$$\text{gr } U(\mathfrak{n}_+) \cong \text{Sym}(\mathfrak{n}_{+, \delta}) \otimes \Lambda(\mathfrak{n}_{+, \bar{\mathbf{i}}}).$$

Put

$$S = \text{Sym}(\mathfrak{n}_{+, \delta}) \otimes \Lambda(\mathfrak{n}_{+, \bar{\mathbf{i}}}).$$

The associated-graded target is $N \text{Bar}_\bullet(k, S, k)$, whereas the associated-graded source has zero differential and is

$$\Lambda(\mathfrak{n}_{+, \delta}) \otimes \text{Sym}(\mathfrak{n}_{+, \bar{\mathbf{i}}}) \cong \text{Sym}_{\text{Ch}(s\text{Vect}_k)}(\mathfrak{n}_+).$$

For one even generator x , the Koszul resolution of k over $k[x]$ shows that the normalized bar has homology $k \oplus k[sx]$, represented by $\mathbf{1}$ and $[x]$. For one odd generator θ , the augmentation ideal of $\Lambda(\theta)$ is $k\theta$ with $\theta^2 = 0$, so its normalized bar differential vanishes and

$$\text{Alt}((s\theta)^r) = r! [\theta | \dots | \theta].$$

Thus both one-generator comparisons are quasi-isomorphisms in characteristic zero. The shuffle Eilenberg–Zilber map and the Künneth theorem identify gr Alt , in each fixed weight, with their finite tensor product. Hence it is a quasi-isomorphism. The finite exhaustive PBW filtration in each weight then proves that Alt is a quasi-isomorphism. The ordinary exterior resolution inside the normalized standard resolution is [4, Chapter XIII, Section 7, Theorem 7.1]; the same filtration argument, with the displayed Koszul signs, gives the super extension. The PBW and chain/cochain conventions are in [4, Chapter XIII, Sections 3 and 8].

The finite-factorization property and finite-dimensional root spaces make every fixed weight a bounded finite-dimensional complex. Euler–Poincaré therefore reduces the character to the underlying superobject $\text{Sym}_{\text{Ch}(s\text{Vect}_k)}(\mathfrak{n}_+)$; the differential d_{CE} , which is generally nonzero, does not change it. An intrinsically even root vector becomes total-odd after suspension and contributes $\mathbf{1} - X^\alpha$. An intrinsically odd root vector becomes total-even and contributes $(\mathbf{1} - X^\alpha)^{-1}$. Multiplication over a homogeneous basis gives the stated exponent and product. For a one-dimensional abelian Lie superalgebra in degree α , with its generator respectively even or odd, these two cases are

$$\mathbf{1} - X^\alpha, \quad \mathbf{1} + X^\alpha + X^{2\alpha} + \dots = (\mathbf{1} - X^\alpha)^{-1},$$

which fixes the homological and intrinsic-parity signs independently. $\square$

Remark 6.3. Theorem 6.2 is an ordinary algebraic result. It has no Ran descent datum. Its product depends only on signed root spaces; a generalized Cartan datum, positive system, Weyl vector, and imaginary-simple-root correction are additional inputs to a superdenominator identity.

Open Problem 6.4 (The carrier-changing comparison). For this problem set $k = \mathbb{C}$; more generally, a symmetric monoidal change-of-coefficients functor must be included among the data. Given an ordered

factorization algebra A , construct a grading-preserving comparison between its realized factorization bar and $\mathcal{B}(\mathfrak{n}_+)$. One sufficient set of data is a strong monoidal functor

$$\Phi : (\text{FactD}(X), \otimes^!, \omega_{\text{Ran}}) \longrightarrow (\widehat{\text{Ch}}_{Q_+}^{\text{pf}}(\text{sVect}_{\mathbb{C}}), \widehat{\otimes}, \mathbb{C}), \quad \Phi(A) \simeq U(\mathfrak{n}_+)$$

as augmented E_1 -algebras, together with preservation of the chosen realization. For a strict augmented dg representative, the comparison must already commute with the length-two bar face induced by multiplication. If $\eta : \Phi(A) \rightarrow U(\mathfrak{n}_+)$ denotes its algebra map and ϕ_2 the tensor comparison of Φ , this equation is

$$\eta \Phi(\mu_A) \phi_2 = \mu_U(\eta \otimes \eta), \quad \varepsilon_U \eta = \Phi(\varepsilon_A),$$

where the unit comparison identifies $\Phi(\omega_{\text{Ran}})$ with $\mathbb{C}$. Here μ_A, μ_U are the two products and $\varepsilon_A, \varepsilon_U$ their augmentations. For a homotopy-coherent comparison these equalities require specified homotopies and their higher compatibility equations. Alternatively, a direct construction must first map both bars to one named root-graded chain category and then give a grading-preserving quasi-isomorphism there.

Conditional Theorem 6.5 (Comparison through a monoidal functor). *Assume the first set of data in Problem 6.4, including the hypotheses of Theorem 5.2. Then there are natural equivalences*

$$\Phi\left(\left|\text{Bar}_{\bullet}^{\mathcal{F}}(\omega_{\text{Ran}}, A, \omega_{\text{Ran}})\right|\right) \simeq |\text{Bar}_{\bullet}(\mathbb{C}, \Phi(A), \mathbb{C})| \simeq \mathcal{B}(\mathfrak{n}_+).$$

Consequently,

$$\chi_{Q_+}^{\text{tot}}\left(\Phi\left(\left|\text{Bar}_{\bullet}^{\mathcal{F}}(\omega_{\text{Ran}}, A, \omega_{\text{Ran}})\right|\right)\right) = \prod_{\alpha \in Q_+ \setminus \{0\}} (1 - X^{\alpha})^{\text{sdimg}_{\alpha}}.$$

Proof. Strong monoidality identifies the unit, multiplication, and augmentation in every simplicial degree. Preservation of realization gives the first equivalence, and the augmented-algebra equivalence gives the second. $\square$

7. FROM A BAR PRODUCT TO A SUPERDENOMINATOR

Let H be a real bilinear space and let $G = G(H, \{h_i\}_{i \in I}, \tau)$ be a Borcherds realization in the sense of Gritsenko–Nikulin, Section 6. Thus $(h_i, h_j) \leq 0$ for $i \neq j$, $2(h_i, h_j)/(h_i, h_i)$ is integral whenever $(h_i, h_i) > 0$, and $(h_i, h_i) \leq 0$ for every odd simple index $i \in \tau$. Assume the hypotheses of their Statement 6.8', so in particular there are no odd real simple roots. Put

$$\tilde{Q} = \bigoplus_{i \in I} \mathbb{Z}r_i, \quad \tilde{Q}_+ = \sum_{i \in I} \mathbb{N}r_i, \quad \varpi(r_i) = h_i,$$

and require the conditions (6.29)–(6.31): every fibre of $\varpi|_{\tilde{Q}_+}$ is finite and

$$\varpi^{-1}(0) \cap \tilde{Q}_+ = \{0\}.$$

Let $Q = \sum_i \mathbb{Z}h_i \subset H$, $Q_+ = \varpi(\tilde{Q}_+)$, and $\Delta_+ = \Delta \cap (Q_+ \setminus \{0\})$. These conditions make Q_+ pointed with the finite-factorization property: every ordered decomposition lifts to an ordered decomposition of one of the finitely many elements in its ϖ -fibre, and an element of the free monoid $\tilde{Q}_+$ has only finitely many such decompositions. Roots $\alpha \in \Delta_+$ below are the Q -valued roots, and G_{α} denotes the total finite-dimensional root space in that degree. Set

$$G_+ = \bigoplus_{\alpha \in \Delta_+} G_{\alpha}, \quad \text{mult } \alpha = \dim G_{\alpha, \bar{0}} - \dim G_{\alpha, \bar{1}}.$$

Let $P \supset Q$ be a W -stable weight lattice containing a Weyl vector ρ_{alg} with

$$(\rho_{\text{alg}}, h_i) = \frac{1}{2}(h_i, h_i).$$

Here W is the Weyl group of the real simple roots and $\varepsilon_W(w) = (-1)^{\ell(w)}$. Write $e(\lambda)e(\mu) = e(\lambda + \mu)$ and $X^\alpha = e(-\alpha)$, and define the shifted completion

$$e(\rho_{\text{alg}})\widehat{\mathbb{Z}[Q_+]} := \prod_{\beta \in Q_+} \mathbb{Z} e(\rho_{\text{alg}} - \beta).$$

Corollary 7.1 (Formal superdenominator comparison). *For*

$$\mathcal{B}(G_+) = N \text{Bar}_\bullet(\mathbb{C}, U(G_+), \mathbb{C}),$$

the unshifted bar character is

$$\chi_{Q_+}^{\text{tot}}(\mathcal{B}(G_+)) = \prod_{\alpha \in \Delta_+} (1 - X^\alpha)^{\text{mult } \alpha}.$$

Its Weyl shift satisfies, coefficientwise in the shifted completion,

$$e(\rho_{\text{alg}})\chi_{Q_+}^{\text{tot}}(\mathcal{B}(G_+)) = e(\rho_{\text{alg}}) \prod_{\alpha \in \Delta_+} (1 - e(-\alpha))^{\text{mult } \alpha}.$$

It is the product side of

$$e(\rho_{\text{alg}}) \prod_{\alpha \in \Delta_+} (1 - e(-\alpha))^{\text{mult } \alpha} = \sum_{w \in W} \varepsilon_W(w) w(S),$$

where

$$S = e(\rho_{\text{alg}}) \sum_s \varepsilon_I(s) e(-\vartheta(s)).$$

Here $s = \mathbf{o}$ is included, with $\varepsilon_I(\mathbf{o}) = 1$. The remaining s range over sums of pairwise perpendicular imaginary simple indices. If s contains $n_{\mathbf{o}}$ even and n_1 odd imaginary simple indices, then $\varepsilon_I(s) = (-1)^{n_{\mathbf{o}}}$; indices outside the set of odd isotropic simple roots are required to be distinct, and $\varepsilon_I(s) = \mathbf{o}$ for an inadmissible sum.

Proof. Theorem 6.2 gives the product on the left. The remaining identity is the superdenominator formula [9, the paragraph before Statement 6.5 and Statement 6.8', with (6.29)–(6.31)]. $\square$

Remark 7.2. For ordinary generalized Kac–Moody algebras the corresponding formula is [3, formula (3), pp. 506–507]. That formula does not contain the super sign or the odd-imaginary-root correction above.

8. THE IGUSA SPECIALIZATION

8.1. The active root cone. Let

$$M_{\mathbf{o}} = \mathbb{Z}f_2 \oplus \mathbb{Z}f_3 \oplus \mathbb{Z}f_{-2}, \quad (f_2, f_{-2}) = -1, \quad (f_3, f_3) = 2,$$

with all other basis pairings zero, and put

$$\delta_1 = 2f_2 - f_3, \quad \delta_2 = 2f_{-2} - f_3, \quad \delta_3 = f_3.$$

Then

$$Q = M_{II} = \mathbb{Z}\delta_1 \oplus \mathbb{Z}\delta_2 \oplus \mathbb{Z}\delta_3, \quad Q_+ = \mathbb{N}\delta_1 + \mathbb{N}\delta_2 + \mathbb{N}\delta_3.$$

Thus $Q_+ \cong \mathbb{N}^3$ has the finite-factorization property. These lattice and chamber data are those of [9, (2.1)–(2.4)].

For $n, \ell, m \in \mathbb{Z}$, define

$$\alpha(n, \ell, m) = 2nf_2 - \ell f_3 + 2mf_{-2}.$$

Its coordinates in Q are

$$\alpha(n, \ell, m) = n\delta_1 + m\delta_2 + (n + m - \ell)\delta_3. \quad (1)$$

The positive triples satisfy $n, m \geq 0$, with arbitrary ℓ if $n > 0$ or $m > 0$, and with $\ell < 0$ if $n = m = 0$ [9, Theorem 4.1 and Remark 4.2].

Let

$$\phi_{0,1}(\tau, z) = \sum_{N \geq 0, \ell \in \mathbb{Z}} f(N, \ell) e^{2\pi i(N\tau + \ell z)}$$

be the weak Jacobi form normalized by $f(0, 0) = 10$ and $f(0, \pm 1) = 1$. It has weight zero and index one, and its support satisfies

$$f(N, \ell) \neq 0 \implies 4N - \ell^2 \geq -1.$$

These assertions follow from [9, (4.11) and the paragraph following it].

Lemma 8.1 (Active positive cone). *If $(n, \ell, m) > 0$ and $f(nm, \ell) \neq 0$, then $\alpha(n, \ell, m) \in Q_+$.*

Proof. If $n + m > 0$ and $\ell \geq n + m + 1$, then

$$\ell^2 \geq (n + m + 1)^2 > 4nm + 1,$$

contrary to the support inequality. Hence $n + m - \ell \geq 0$. If $n = m = 0$, positivity and the support inequality force $\ell = -1$. Equation (1) gives the result. $\square$

The restriction to active roots is essential. The raw positive triples have infinitely many decompositions

$$(1, 0, 0) = (1, j, 0) + (0, -j, 0), \quad j \geq 1.$$

Both summands are simultaneously active only for $j = 1$. At the boundary $n = m = 0$, the cone selects $\delta_3 = \alpha(0, -1, 0)$. Under the analytic substitution below its factor is $1 - y^{-2}$; reversing the root changes it by the exact monomial factor

$$1 - y^2 = -y^2(1 - y^{-2}).$$

Gritsenko and Nikulin construct a generalized Kac–Moody superalgebra $\mathfrak{g}_{\Delta_3}$ with finite-dimensional root spaces and

$$\text{sdim}(\mathfrak{g}_{\Delta_3})_{\alpha(n, \ell, m)} = f(nm, \ell)$$

[9, Section 3 and (3.4)]. The finite-fibre conditions ensuring finite-dimensional projected root spaces are [9, (6.29)–(6.31)]. Define

$$\mathfrak{n}_{\Delta_3, +} = \bigoplus_{\alpha \in \Delta_+} (\mathfrak{g}_{\Delta_3})_{\alpha}, \quad \mathcal{B}_{\Delta_3} = N \text{Bar}_{\bullet}(\mathbb{C}, U(\mathfrak{n}_{\Delta_3, +}), \mathbb{C}).$$

By Lemma 8.1, this is weightwise perfect in $\text{Ch}_{\geq 0}(\text{sVect}_{\mathbb{C}}^{Q_+})$.

8.2. The Weyl shift and the formal Fourier map. The geometric Weyl vector is

$$\rho_{\text{geo}} = \frac{1}{2}(\delta_1 + \delta_2 + \delta_3), \quad (\rho_{\text{geo}}, \delta_i) = -1.$$

The algebraic convention in the superdenominator formula is

$$\rho_{\text{alg}} = -\rho_{\text{geo}}, \quad (\rho_{\text{alg}}, \delta_i) = 1.$$

These conventions are respectively [9, (2.3)–(2.4), (3.2)–(3.3)] and the paragraph preceding [9, Statement 6.5, together with Statement 6.8']. In particular,

$$e(\rho_{\text{alg}}) = e(-\rho_{\text{geo}}) = X^{\rho_{\text{geo}}}, \quad X^{\alpha} := e(-\alpha).$$

Since ρ_{geo} lies in the weight lattice but not in Q , this expression belongs to the shifted module

$$X^{\rho_{\text{geo}}} \widehat{\mathbb{Z}[Q_+]}. \quad \square$$

Let

$$\widehat{\mathcal{F}}_{Q_+} = \mathbb{Z}[[u, v, w]]$$

with its (u, v, w) -adic topology. Let $\widehat{\omega\mathcal{F}_{Q_+}}$ be the free rank-one shifted module obtained in the enlarged lattice algebra by imposing $\omega^2 = uvw$. There is a continuous formal isomorphism

$$\mathcal{F}_2 : \widehat{\mathbb{Z}[Q_+]} \longrightarrow \widehat{\mathcal{F}_{Q_+}}, \quad X^{\delta_1} \mapsto u, \quad X^{\delta_2} \mapsto v, \quad X^{\delta_3} \mapsto w,$$

and a shifted extension

$$\mathcal{F}_2^\rho : X^{\rho_{\text{geo}}} \widehat{\mathbb{Z}[Q_+]} \longrightarrow \widehat{\omega\mathcal{F}_{Q_+}}, \quad \mathcal{F}_2^\rho(X^{\rho_{\text{geo}}} F) = \omega \mathcal{F}_2(F).$$

Proposition 8.2 (Formal Igusa character). *One has*

$$\mathcal{F}_2^\rho \left(X^{\rho_{\text{geo}}} \chi_{Q_+}^{\text{tot}}(\mathcal{B}_{\Delta_i}) \right) = \omega \prod_{\substack{(n,\ell,m) > 0 \\ f(nm,\ell) \neq 0}} \left(1 - u^n v^m w^{n+m-\ell} \right)^{f(nm,\ell)}$$

in $\widehat{\omega\mathcal{F}_{Q_+}}$.

Proof. Apply Theorem 6.2 and the signed multiplicity formula (3.4) of [9]. Equation (1) gives the exponent of $u^n v^m w^{n+m-\ell}$, while the Weyl-vector convention gives the factor ω . $\square$

Remark 8.3. Proposition 8.2 is a formal equality in a shifted cone module. Neither the scalar 64 nor analytic convergence can be recovered from the exponent table.

8.3. Analytic realization at the cusp. For

$$Z = \begin{pmatrix} z_1 & z_2 \\ z_2 & z_3 \end{pmatrix} \in \mathbb{H}_2, \quad q = e^{2\pi i z_1}, \quad y = e^{2\pi i z_2}, \quad p = e^{2\pi i z_3},$$

set

$$u_Z = q^2 y^2, \quad v_Z = p^2 y^2, \quad w_Z = y^{-2}, \quad \omega_Z = q y p.$$

Then

$$\omega_Z^2 = u_Z v_Z w_Z, \quad u_Z^n v_Z^m w_Z^{n+m-\ell} = q^{2n} y^{2\ell} p^{2m}.$$

Writing $y_j = \text{Im } z_j$, define

$$U_T = \{Z \in \mathbb{H}_2 : -y_2 > T, \quad y_1 + y_2 > T, \quad y_3 + y_2 > T\}.$$

For $\zeta = z_3 f_2 + z_2 f_3 + z_1 f_{-2}$, these inequalities are equivalent to $(\text{Im } \zeta, \delta_i) < -2T$. In particular,

$$|u_Z|, |v_Z|, |w_Z| < e^{-4\pi T}.$$

Let $\widehat{\mathcal{F}_{Q_+}^{\text{abs}}}(U_T)$ consist of the series $F = \sum A_{a,b,c} u^a v^b w^c$ for which

$$\|F\|_K = \sum_{a,b,c \geq 0} |A_{a,b,c}| \sup_{Z \in K} |u_Z^a v_Z^b w_Z^c| < \infty$$

for every compact $K \Subset U_T$. Endow this algebra with the seminorms $\|\cdot\|_K$, and $\mathcal{O}(U_T)$ with the compact-open topology. Termwise summation defines continuous maps

$$\text{an}_{U_T} : \widehat{\mathcal{F}_{Q_+}^{\text{abs}}}(U_T) \longrightarrow \mathcal{O}(U_T),$$

$$\text{an}_{U_T}^\rho : \widehat{\omega\mathcal{F}_{Q_+}^{\text{abs}}}(U_T) \longrightarrow \mathcal{O}(U_T), \quad \omega F \mapsto (q y p) \text{an}_{U_T}(F).$$

There is no such termwise realization on the whole adic completion: $\sum_{r \geq 0} r! u^r$ has radius of convergence zero.

For an active triple put

$$d = n + m + (n + m - \ell).$$

The support inequality gives

$$\sqrt{\max(0, 4nm - \ell^2)} \leq n + m \leq d,$$

and only $O(d^2)$ triples have a fixed value of d . The coefficient estimate in the proof of [9, Theorem 4.1], together with

$$\left| u_Z^n v_Z^m w_Z^{n+m-\ell} \right| \leq e^{-4\pi T d},$$

therefore implies that, for sufficiently large T ,

$$\sum_{\substack{(n,\ell,m) > 0 \\ f(nm,\ell) \neq 0}} |f(nm,\ell)| \sup_{Z \in K} \frac{|q^{2n} y^{2\ell} p^{2m}|}{1 - |q^{2n} y^{2\ell} p^{2m}|} < \infty$$

for every compact $K \subseteq U_T$. Consequently the formal product in Proposition 8.2 belongs to $\widehat{\omega_{\mathcal{F}_{Q_+}^{\text{abs}}}}(U_T)$: for a factor with exponent a , its absolute binomial norm is bounded by $\exp(|a|r/(1-r))$ when the monomial has modulus at most $r < 1$.

Let Δ_5 be the product of the ten even genus-two theta constants, and put $D_5 = 64^{-1}\Delta_5$.

Theorem 8.4 (Igusa realization). *For sufficiently large T ,*

$$\begin{aligned} \text{an}_{U_T}^{\rho} \left(\mathcal{F}_2^{\rho} \left(X^{\rho_{\text{geo}}} \chi_{Q_+}^{\text{tot}}(\mathcal{B}_{\Delta_5}) \right) \right) \\ = q\gamma p \prod_{\substack{(n,\ell,m) > 0 \\ f(nm,\ell) \neq 0}} \left(1 - q^{2n} y^{2\ell} p^{2m} \right)^{f(nm,\ell)} \\ = D_5(2Z) = 64^{-1}\Delta_5(2Z) \end{aligned}$$

as holomorphic functions on U_T .

Proof. The product formula and its convergence are [9, Theorem 4.1 and its proof, Remark 4.2], evaluated at $2Z$. The normalization $D_5 = 64^{-1}\Delta_5$ follows from the ten-theta product and its leading Fourier coefficient [9, (1.2), (1.6)–(1.7)]. The identification with the superdenominator is [9, Proposition 3.1]; the scale $Z \mapsto 2Z$ is explained in [9, Remark 3.2]. $\square$

Remark 8.5 (Automorphic descent). The form Δ_5 satisfies

$$\Delta_5(M\langle Z \rangle) = v(M) \det(CZ + D)^5 \Delta_5(Z), \quad M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in \text{Sp}_4(\mathbb{Z}),$$

where v is the nontrivial multiplier of [9, (1.3)–(1.5)]. Thus it is a section of the weight-five automorphic line with multiplier v .

Let

$$g_2 = \begin{pmatrix} 2I_2 & 0 \\ 0 & I_2 \end{pmatrix} \in \text{GSp}_4(\mathbb{Q}), \quad \Gamma^{(2)} = g_2^{-1} \text{Sp}_4(\mathbb{Z}) g_2.$$

For $\gamma \in \Gamma^{(2)}$, put $M = g_2 \gamma g_2^{-1}$. Then

$$D_5(2\gamma\langle Z \rangle) = v(M) \det(C_M(2Z) + D_M)^5 D_5(2Z).$$

Accordingly, $D_5(2Z)$ is the pullback of the weight-five automorphic section along the scaled cusp embedding. The formal bar character has no equivariance datum of this kind; automorphic descent is supplied by the separate Gritsenko–Nikulin theorem.

9. THE POSITIVE WITT ALGEBRA

Let

$$L_1^{\text{poly}} = \bigoplus_{j \geq 1} \mathbb{C} e_j, \quad \widehat{L}_1 = \prod_{j \geq 1} \mathbb{C} e_j, \quad [e_i, e_j] = (j - i) e_{i+j}.$$

Equip $\widehat{L}_1$ with the filtration topology

$$F^N \widehat{L}_1 = \prod_{j \geq N} \mathbb{C} e_j.$$

For every n, w , evaluation gives a perfect pairing

$$C_n^{\text{CE}}(L_1^{\text{poly}}; \mathbb{C})_w \otimes C_{\text{cont}}^n(\widehat{L}_1; \mathbb{C})_w \longrightarrow \mathbb{C}.$$

Indeed, only finitely many strictly increasing positive indices sum to w . The chain and cochain differentials are adjoint, hence

$$H_n^{\text{CE}}(L_1^{\text{poly}}; \mathbb{C})_w \cong H_{\text{cont}}^n(\widehat{L}_1; \mathbb{C})_w^\vee. \quad (2)$$

The chain–cochain pairing is the one in [8, equations (2)–(3)]. The bar and CE chain constructions are covariant; continuous cochains are contravariant. The duality in (2) is the variance-reversing bridge.

Theorem 9.1 (Positive Witt bar homology). *For $n \geq 1$,*

$$\dim H_n \mathcal{B}(L_1^{\text{poly}})_w = \begin{cases} 1, & w = (3n^2 - n)/2 \text{ or } w = (3n^2 + n)/2, \\ 0, & \text{otherwise.} \end{cases}$$

Thus $\dim H_n \mathcal{B}(L_1^{\text{poly}}) = 2$ for every $n \geq 1$.

Proof. Theorem 6.2 identifies the normalized enveloping-algebra bar with the CE chain complex, weight by weight. Equation (2) identifies its homology with the dual of continuous cohomology. Goncharova’s Main Theorem and Example 1.1 give precisely the two displayed weights and the vanishing elsewhere [8, Sections 1.1–1.3, pp. 8–10]. $\square$

The smallest weight calculation is

$$[L_1^{\text{poly}}, L_1^{\text{poly}}] = \text{span}\{e_j : j \geq 3\},$$

so H_1 has weights 1 and 2.

Corollary 9.2 (Pentagonal character). *In $\mathbb{Z}[[q]]$,*

$$\begin{aligned} \chi_q^{\text{tot}} \mathcal{B}(L_1^{\text{poly}}) &= \prod_{m \geq 1} (1 - q^m) \\ &= 1 + \sum_{n \geq 1} (-1)^n \left(q^{(3n^2 - n)/2} + q^{(3n^2 + n)/2} \right). \end{aligned}$$

Proof. The first equality is Theorem 6.2. The second is Euler’s pentagonal number theorem [6, Section 27.14(ii), equations (27.14.4)–(27.14.5)]. $\square$

Remark 9.3. The product corroborates the weights in Theorem 9.1; it does not prove the vanishing statement. Euler characteristic cannot see a direct sum of one-dimensional zero-differential chain complexes in two adjacent degrees at the same weight. The absence of additional classes comes from Goncharova’s theorem.

10. MOTZKIN–RIORDAN IDENTITIES AND ORDINARY CARRIERS

Let M_n count Motzkin paths of length n , and let R_n count those with no horizontal step at height zero. Put

$$M_0 = R_0 = 1, \quad R_1 = 0,$$

and define

$$M(z) = \sum_{n \geq 0} M_n z^n, \quad R(z) = \sum_{n \geq 0} R_n z^n.$$

Proposition 10.1 (The Motzkin–Riordan relation). *The generating functions satisfy*

$$M = 1 + zM + z^2M^2, \quad R = 1 + z^2MR.$$

Consequently,

$$R = \frac{M}{1 + zM}, \quad M = R + \frac{R - 1}{z},$$

and, for every $n \geq 0$,

$$M_n = R_n + R_{n+1}, \quad M_{n+1} - M_n = R_{n+2} - R_n.$$

Moreover,

$$M(z) = \frac{1 - z - \sqrt{1 - 2z - 3z^2}}{2z^2}, \quad R(z) = \frac{1 + z - \sqrt{1 - 2z - 3z^2}}{2z(1 + z)}.$$

Proof. A nonempty Motzkin path is either a horizontal step followed by a Motzkin path, or has a unique first-return decomposition $UPDQ$, with P, Q Motzkin paths. This gives the first equation for M . A nonempty Riordan path has the same form $UPDQ$, now with P Motzkin and Q Riordan, which gives the equation for R . Algebraic elimination yields the two rational relations and coefficient comparison gives the identities. Solving the quadratic equation for the branch with constant term 1 gives the displayed closed forms. These identities and generating equations also appear in [2, pp. 87, 89, 100–101]; the path convention is [5, Section 2]. $\square$

Remark 10.2. Splitting a Motzkin path according to whether its first step is horizontal does not prove $M_n = R_n + R_{n+1}$. At length three the paths

$$HHH, \quad HUD, \quad UDH, \quad UHD$$

split as $2 + 2$, whereas $R_3 + R_4 = 1 + 3$. The common square-root discriminant follows from Proposition 10.1. This scalar identity neither constructs nor rules out a representation-theoretic or Drinfeld–Sokolov comparison.

Proposition 10.3 (The ordinary $\mathfrak{sl}_2$ calculation). *Let k have characteristic zero and give $\mathfrak{sl}_2(k)$ the trivial module k . Then*

$$H^\bullet(\mathfrak{sl}_2; k) \cong \Lambda(c_3), \quad |c_3| = 3,$$

and the CE cochain dimensions are $(1, 3, 3, 1)$. In contrast, every positive simplicial degree of

$$N \operatorname{Bar}_\bullet(k, U(\mathfrak{sl}_2), k)$$

is infinite-dimensional.

Proof. Use $[h, e] = 2e$, $[h, f] = -2f$, $[e, f] = h$. On the dual basis,

$$de^\vee = 2e^\vee \wedge h^\vee, \quad df^\vee = -2f^\vee \wedge h^\vee, \quad dh^\vee = -e^\vee \wedge f^\vee.$$

Thus $d : C^1 \rightarrow C^2$ is an isomorphism and $d : C^2 \rightarrow C^3$ is zero. The class

$$c_3 = [e^\vee \wedge f^\vee \wedge h^\vee]$$

generates H^3 , which proves the cohomology statement directly.

For the normalized associative bar,

$$(N \operatorname{Bar}_\bullet(k, U(\mathfrak{sl}_2), k))_r \cong \overline{U(\mathfrak{sl}_2)}^{\otimes r}.$$

PBW gives linearly independent elements $e, e^2, e^3, \dots$ in the augmentation ideal. Length one is already infinite-dimensional, and hence so is every positive length. $\square$

Remark 10.4 (Ordinary and chiral carriers). For the standard Virasoro bracket, $e_n \mapsto -L_n$ identifies L_1^{poly} with the positive-mode Lie subalgebra, because the central cocycle vanishes on positive indices. This

inclusion does not identify the ordinary enveloping-algebra bar with a chiral bar of a Virasoro vacuum object. Such a comparison must specify the central charge, vacuum object, completion, homological and conformal gradings, and a map relating the two complexes.

Likewise, Proposition 10.3 has no implication for an affine or chiral $\widehat{\mathfrak{sl}}_2$ object without a specified level, vacuum module, completion, grading, and comparison. A grading-preserving quasi-isomorphism would suffice. A spectral-sequence argument additionally requires an identified abutment, strong convergence, and control of all relevant differentials.

For the ordinary carriers, consider the two proposed scalar assignments

$$\dim H_n \mathcal{B}(L_1^{\text{poly}}) = M_{n+1} - M_n, \quad \dim H^n(\mathfrak{sl}_2; k) = R_{n+3}.$$

Their first discrepancies are decisive:

$$M_2 - M_1 = 1 \neq 2 = \dim H_1 \mathcal{B}(L_1^{\text{poly}}), \quad R_4 = 3 \neq 0 = \dim H^1(\mathfrak{sl}_2; k).$$

A finite total Betti number in each degree is a meaningful invariant; the issue is not its scalar type but the absence of a comparison with the proposed chiral or affine carrier.

II. FROM OPERATIONS TO THE CUSP PRODUCT

The algebraic product follows from a chain comparison and finite weightwise Euler characteristics. Its analytic realization then uses the specified Fourier map and cusp domain. The following functors make these successive operations explicit and identify where a geometric factorization comparison would enter.

Fix a finite-factorization monoid Q_+ . Let $\text{LieSup}_k^{Q_+, \text{fd}, +}$ denote the category of positively Q_+ -graded Lie superalgebras with finite-dimensional pieces. Let $\text{Alg}_{E_1}^{\text{aug}, Q_+, \text{fd}, +}$ denote augmented Q_+ -graded superalgebras B with $B_0 = k$, augmentation ideal supported away from zero, and finite-dimensional homogeneous pieces. Enveloping algebra and normalized bar are functors

$$\text{LieSup}_k^{Q_+, \text{fd}, +} \xrightarrow{U} \text{Alg}_{E_1}^{\text{aug}, Q_+, \text{fd}, +}(\text{sVect}_k) \xrightarrow{N \text{ Bar}_\bullet(k, -; k)} \widehat{\text{Ch}}_{Q_+}^{\text{pf}}(\text{sVect}_k).$$

Finite factorization makes the last functor land in its stated target. Theorem 6.2 supplies the natural bar–CE quasi-isomorphism on the first category.

For the geometric branch, the typed arrows are

$$\text{Alg}_{E_1}^{\text{aug}}(\text{FactD}(X), \otimes^!) \xrightarrow{\text{Bar}_\bullet^{\mathcal{F}}} \text{Fun}(\Delta^{\text{op}}, \text{FactD}(X)) \xrightarrow{[-]} \text{FactD}(X) \xrightarrow{-\Phi} \widehat{\text{Ch}}_{Q_+}^{\text{pf}}(\text{sVect}_{\mathbb{C}}).$$

The upper realization arrow is conditional on Theorem 5.2; Φ is the open construction in Problem 6.4. Neither dashed arrow is used in the algebraic theorem.

For an object C of the completed chain category, passage to its coefficientwise class and then to its total character is the typed sequence

$$\text{Obj}(\widehat{\text{Ch}}_{Q_+}^{\text{pf}}(\text{sVect}_k)) / \simeq \xrightarrow{[-]} \widehat{K}_{\text{os}, s}^{Q_+}(k) \xrightarrow{\chi_{Q_+}^{\text{tot}}} \widehat{\mathbb{Z}[Q_+]}. \quad \square$$

The second arrow is a group homomorphism, not a chain functor. The Igusa specialization continues from the final ring by the shifted formal map $\mathcal{F}_2^{\rho}$, then by the restricted analytic realization on U_T . Neither map constructs Φ .

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